

PASSWORD STRENGTH CLASSIFICATION

23CSA391A

Final Report

Submitted by

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Abstract

Passwords continue to be the primary method of authentication in digital systems, yet weak and predictable passwords remain a major cause of security breaches. Traditional strength meters rely only on simple checks such as length or character variety, which often fail to capture deeper structural weaknesses. This project, *Password Strength Classification*, proposes an enhanced, hybrid password evaluation system that combines rule-based analysis with entropy estimation, pattern identification, and explainable weakness detection.

The developed system evaluates passwords based on multiple security parameters, including character diversity, entropy (in bits), dictionary similarity, sequential patterns, repeated characters, and attacker speed assumptions. A graphical user interface (GUI) was built using Python and Tkinter to provide real-time feedback, visual strength indicators, actionable suggestions, and explainable weak segments. Additional features such as entropy-based crack-time estimation, adjustable attacker speed ($1e3$ – $1e12$ guesses per second), session history logging, CSV export, and improved password generation enhance both usability and security awareness.

By integrating technical analysis with user-friendly explanations, the system helps users understand *why* a password is weak and *how* it can be improved. The project demonstrates a practical and effective approach to strengthening everyday password practices and highlights the importance of combining cybersecurity principles with accessible design. The results show that the model offers more meaningful feedback compared to traditional meters, making it a valuable tool for educational and security-improvement purposes.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
GUI	Graphical User Interface
CSV	Comma Separated Values
BCA	Bachelor of Computer Applications
UI	User Interface
API	Application Programming Interface
CPU	Central Processing Unit
GPU	Graphics Processing Unit
UTF	Unicode Transformation Format
OS	Operating System
sec	Seconds

1. INTRODUCTION:

1.1 BACKGROUND OF THE STUDY:

In the present digital world, information security has become a critical concern due to the rapid growth of internet-based services and applications. Almost every digital platform, including email systems, social networking sites, banking applications, educational portals, and enterprise systems, relies on user authentication mechanisms to protect sensitive information. Among various authentication methods, password-based authentication remains the most widely used due to its simplicity, low cost, and ease of implementation.

Despite its popularity, password-based security continues to be vulnerable because many users choose weak and predictable passwords. Users often prefer passwords that are easy to remember, such as names, dates of birth, phone numbers, or simple numeric and keyboard patterns. These predictable choices significantly reduce the effectiveness of passwords and make systems vulnerable to common cyberattacks such as brute-force attacks, dictionary attacks, and credential stuffing.

To address this issue, many applications employ password strength checkers to guide users in creating secure passwords. Traditional password strength meters usually evaluate passwords based on simple criteria such as minimum length, inclusion of uppercase letters, numbers, or special characters. While these checks provide basic guidance, they often fail to detect deeper structural weaknesses such as repeated patterns, dictionary words, or predictable sequences. As a result, users may receive misleading feedback about the actual security of their passwords.

This project focuses on developing an enhanced Password Strength Classification System that evaluates passwords more accurately by incorporating entropy calculation, pattern detection, and explainable feedback. By providing clear strength indicators and meaningful suggestions, the system aims to improve user awareness and promote better password practices.

1.2 PROBLEM STATEMENT

Despite increased awareness of cybersecurity threats, weak passwords continue to be a major cause of security breaches across various digital platforms. Existing password strength meters often provide only a general strength label, such as “weak” or “strong,” without explaining the underlying reasons for the classification. This lack of transparency prevents users from understanding how their passwords can be improved.

The major issues with existing password strength evaluation systems include:

- Dependence on simple rule-based checks that do not reflect real-world attack scenarios.
- Inability to detect predictable patterns such as dictionary words, sequences, or repeated characters.
- Absence of meaningful metrics such as entropy and estimated password crack time.
- Lack of explainable feedback that educates users about password weaknesses. Limited user interaction and poor visualization of password security levels.

As a result, users may develop a false sense of security and continue using passwords that are vulnerable to attacks. Therefore, there is a need for a more advanced password strength classification system that evaluates passwords comprehensively and provides clear, actionable feedback to users.

1.3 OBJECTIVES OF THE PROJECT

The main objective of this project is to design and develop a password strength classification system that evaluates the security of passwords in a more accurate and meaningful manner.

The specific objectives of the project are:

- To analyze passwords based on multiple security factors such as length, character diversity, entropy, and pattern structure.

- To identify weak patterns in passwords, including dictionary words, numeric sequences, repeated characters, and predictable combinations.
- To estimate the time required to crack a password by considering entropy and attacker speed assumptions.
- To provide clear, explainable feedback to users about why a password is weak or strong.
- To suggest improved password combinations that are both secure and user-friendly.
- To design an interactive graphical user interface for real-time password strength evaluation.
- To enhance user awareness about good password practices and cybersecurity principles.

1.4 SCOPE OF THE PROJECT

The scope of this project is focused on the development of a password strength classification system for educational and security-awareness purposes. The system evaluates passwords locally and provides real-time feedback without storing or transmitting user-entered passwords, ensuring privacy and ethical usage.

The project covers password analysis based on parameters such as password length, character variety, entropy, dictionary similarity, sequential patterns, and repeated characters. It also includes features such as estimated crack time calculation, attacker speed variation, explainable weak-segment identification, session history logging, and CSV export of analysis results.

The project does not include real-time password cracking, hacking tools, or integration with live authentication systems. Machine learning–based password classification and browser extension development are beyond the scope of the current implementation but can be explored in future work.

Despite these limitations, the project provides a strong foundation for understanding password security principles and demonstrates how effective password strength classification can reduce cybersecurity risks.

2. LITERATURE REVIEW

Introduction

A literature review provides an overview of existing research, systems, and techniques related to password security. Several approaches have been proposed to evaluate password strength and reduce unauthorized access. This chapter reviews commonly used password strength evaluation methods, highlights their limitations, and explains how the proposed system addresses these gaps.

Traditional Password Strength Meters

Traditional password strength meters are commonly integrated into applications and websites to guide users while creating passwords. These meters typically evaluate passwords based on basic rules such as minimum length, inclusion of uppercase and lowercase letters, numbers, and special characters. Although such systems are easy to implement, they often fail to detect predictable patterns and real-world vulnerabilities.

Dictionary-Based and Rule-Based Approaches

Dictionary-based approaches compare passwords against lists of commonly used or leaked passwords. Rule-based methods extend this approach by identifying patterns such as repeated characters, numeric sequences, and keyboard patterns. While these techniques improve detection of weak passwords, they often lack detailed explanation and user-friendly feedback.

Entropy-Based Password Evaluation

Entropy-based evaluation measures password strength mathematically by estimating unpredictability. Higher entropy values indicate stronger passwords that require more guesses to crack. However, entropy alone does not always reflect human password behavior, as some high-entropy passwords may still follow predictable structures.

Limitations of Existing Systems

Despite improvements, many existing password strength checkers rely heavily on basic rules or single metrics. They often do not consider attacker capabilities, crack time estimation, or explainable feedback. As a result, users may not fully understand why a password is weak or how to improve it.

Research Gap and Justification of the Proposed System

The review of existing systems reveals the need for a comprehensive password strength classification approach that combines rule-based analysis, entropy calculation, pattern detection, and user-friendly explanations. The proposed system addresses these gaps by providing explainable weak-segment identification, crack time estimation based on attacker speed, session history tracking, and visual feedback through a graphical user interface.

3. SYSTEM ANALYSIS

System analysis involves understanding the existing methods used to solve a problem, identifying their limitations, and proposing an improved solution. This chapter discusses the current password strength evaluation systems, the proposed system, and the functional and non-functional requirements of the project.

3.1 EXISTING SYSTEM

In existing systems, password strength evaluation is generally performed using simple rule-based mechanisms. These systems check whether a password satisfies basic conditions such as minimum length, presence of uppercase letters, lowercase letters, numbers, and special characters. Many commonly used applications and websites rely on these basic checks to classify passwords as weak, medium, or strong.

While such systems are easy to implement and provide quick feedback, they suffer from several drawbacks. They do not consider real-world attack scenarios and often fail to detect predictable password structures. Passwords like “Password@123” or “User2024” may pass basic checks but remain highly vulnerable to dictionary and brute-force attacks. Moreover, existing systems usually do not explain why a password is weak or how quickly it can be compromised.

3.2 PROPOSED SYSTEM

The proposed system introduces an enhanced Password Strength Classification approach that evaluates passwords more comprehensively. Unlike traditional systems, this project analyzes passwords using multiple parameters, including entropy calculation, pattern detection, dictionary similarity, and attacker speed assumptions.

The system identifies weak segments within a password and provides explainable feedback to help users understand the reasons for vulnerability. It also estimates the time required to crack a password by considering different attacker capabilities. A graphical user interface (GUI) is provided to display password strength, suggestions, crack-time estimation, session history, and exportable results.

The proposed system aims to improve both security and user awareness by offering meaningful and actionable insights rather than simple strength labels.

Table 3.1: Comparison of Existing and Proposed System

Feature	Existing System	Proposed System
Password Evaluation	Basic rule-based checks	Advanced multi-parameter analysis
Strength Criteria	Length and character presence	Entropy, patterns, attacker speed
Pattern Detection	Not supported	Detects sequences, dictionary words
Crack Time Estimation	Not available	Estimated using attacker speed
User Feedback	Simple strength label	Explainable weak segments & suggestions
Password Suggestions	Limited or none	Actionable improvement suggestions
Session History	Not maintained	Stored in masked format
Export Facility	Not available	CSV export supported
User Awareness	Low	High

3.3 FUNCTIONAL REQUIREMENTS

Functional requirements define what the system is expected to do. The functional requirements of the proposed system are listed below:

- The system shall accept password input from the user.
- The system shall evaluate password strength based on predefined security rules.
- The system shall calculate password entropy.

- The system shall detect weak patterns such as dictionary words, sequences, and repeated characters.
- The system shall estimate password crack time based on attacker speed.
- The system shall classify passwords into different strength levels.
- The system shall display password strength visually using a GUI.
- The system shall provide suggestions for improving weak passwords.
- The system shall maintain session-based history of password evaluations.
- The system shall allow exporting analysis results in CSV format.

Table 3.2: Functional Requirements

Requirement ID	Description
FR1	The system shall accept password input from the user
FR2	The system shall evaluate password strength
FR3	The system shall calculate password entropy
FR4	The system shall detect weak password patterns
FR5	The system shall estimate password crack time
FR6	The system shall classify password strength
FR7	The system shall display results through GUI
FR8	The system shall provide improvement suggestions
FR9	The system shall maintain session history
FR10	The system shall export results in CSV format

3.4 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements specify the quality and performance characteristics of the system. The non-functional requirements of the proposed system are as follows:

- Usability: The system should be user-friendly and easy to understand.
- Performance: Password evaluation should be performed in real time.
- Security: User-entered passwords should not be stored or transmitted.
- Reliability: The system should produce consistent results for repeated inputs.
- Portability: The system should run on standard personal computers.
- Maintainability: The system should be easy to modify and extend.
- Scalability: The system design should allow future enhancements.

Table 3.3: Non-Functional Requirements

Requirement Type	Description
Usability	The system should be easy to use and understand
Performance	Password evaluation should be real-time
Security	Passwords should not be stored or exposed
Reliability	The system should give consistent results
Portability	The system should run on standard PCs
Maintainability	The system should be easy to update
Scalability	The system should support future enhancements

4. METHODOLOGY AND SYSTEM DESIGN

This chapter explains the methodology adopted for the development of the Password Strength Classification system and presents the overall system design. It describes the architectural structure, workflow of password evaluation, and user interaction with the system.

4.1 SYSTEM ARCHITECTURE

The system architecture of the Password Strength Classification project is designed using a modular approach to ensure clarity, scalability, and ease of maintenance. The architecture defines how different components of the system interact with each other to evaluate and classify password strength efficiently.

The system primarily consists of four major modules: User Interface Module, Password Analysis Module, Evaluation Engine, and Output Module.

The User Interface Module serves as the interaction point between the user and the system. It is developed using Python's Tkinter library and allows users to enter passwords, view strength classifications, receive suggestions, and analyze results in real time.

The Password Analysis Module processes the input password and performs various checks such as password length validation, character diversity analysis, detection of repeated characters, identification of sequential patterns, and matching against commonly used or weak password patterns. This module ensures that predictable and vulnerable password structures are accurately detected.

The Evaluation Engine is the core component of the system. It calculates password entropy to measure unpredictability and estimates the time required to crack the password by considering different attacker speed assumptions.

Based on the combined results of analysis and entropy calculation, the engine classifies the password into appropriate strength levels.

The Output Module presents the evaluation results to the user in an understandable and visual format. It displays password strength labels, progress indicators, crack-time estimation, weak segment explanations, improvement suggestions, and session history. The module also provides options to export results in CSV format for further analysis.

This modular architecture ensures that each component performs a specific function, making the system reliable, extensible, and easy to upgrade with additional features in the future.

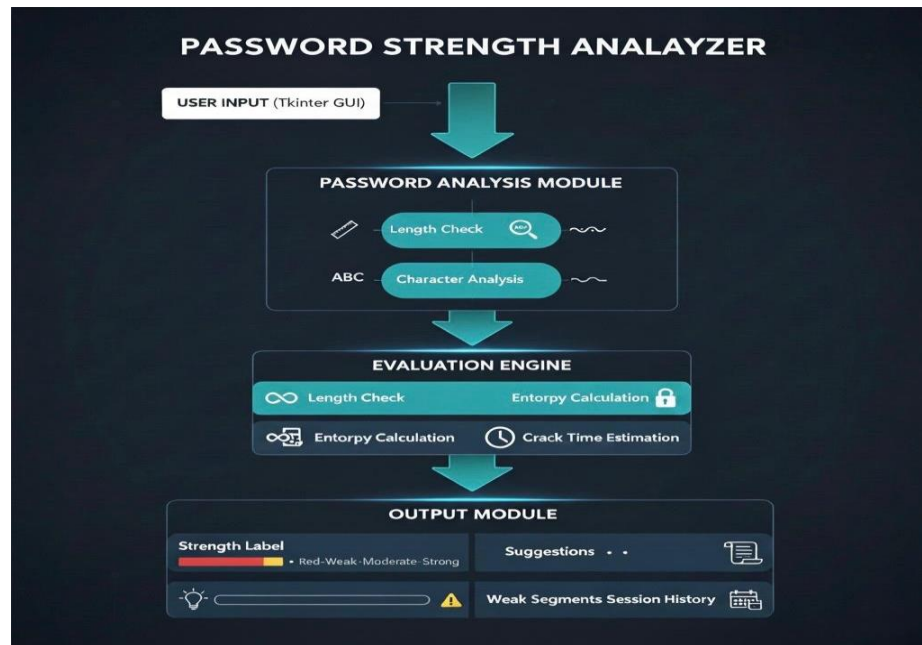


Figure 4.1: System Architecture of Password Strength Classification

Figure 4.1 shows the overall system architecture of the Password Strength Classification System. The user provides input through a graphical user interface developed using Tkinter. The entered password is processed by the Password Analysis Module, which performs length checking and character analysis. The Evaluation Engine then calculates password entropy and estimates the crack time. Finally, the Output Module displays the password strength label, weak segments, improvement suggestions, and maintains session history. This architecture clearly illustrates the modular design and data flow of the system.

4.2 FLOW CHART

The flowchart illustrates the step-by-step process followed by the Password Strength Classification system to evaluate and classify a password. It represents the logical flow of operations from user input to the final output generation.

The process begins when the user enters a password through the graphical user interface. The system first validates the input and checks whether the password is empty. If the input is invalid, the system prompts the user to enter a valid password.

Once a valid password is received, the system performs password analysis, which includes length verification, character type analysis, and pattern detection. After extracting these features, the system calculates the password entropy to determine its unpredictability.

Based on the entropy value and predefined rules, the system estimates the time required to crack the password by considering attacker speed assumptions. The evaluated results are then used to classify the password into appropriate strength levels such as Weak, Moderate, or Strong.

Finally, the system displays the results to the user, including the strength label, visual indicators, weak segment explanations, actionable suggestions, and session history. The user may continue testing additional passwords or exit the system.

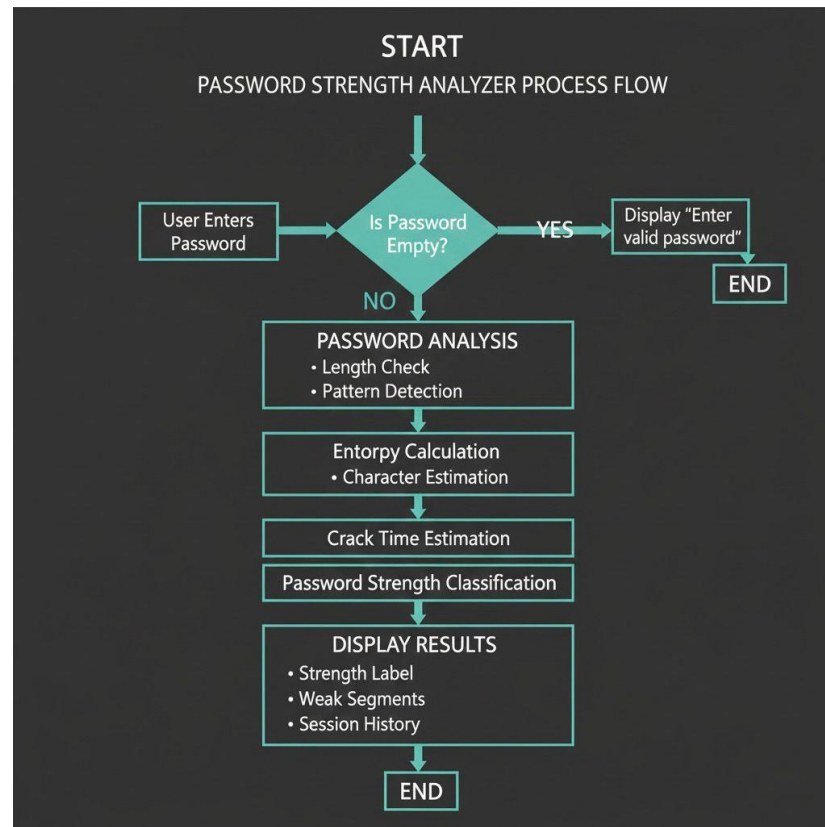


Figure 4.2: Flowchart of Password Strength Evaluation Process

Figure 4.2 represents the step-by-step flow of the Password Strength Classification process. The flow begins when the user enters a password. The system first checks whether the password is empty. If the password is invalid, an error message is displayed. Otherwise, the system proceeds with password analysis, entropy calculation, crack time estimation, and strength classification. The final results, including strength label, weak segments, and session history, are then displayed to the user before the process ends.

4.3 USE CASE DIAGRAM

The use case diagram represents the interaction between the user and the Password Strength Classification system. It illustrates the various functionalities provided by the system and how the user accesses these features.

In this system, the User is the primary actor. The user interacts with the system by entering a password and requesting its strength evaluation. The system processes the password internally and provides meaningful feedback such as strength classification, weak segment identification, and security suggestions.

The use case diagram helps in understanding the functional requirements of the system from the user's perspective and ensures that all major operations of the system are clearly defined and accessible.

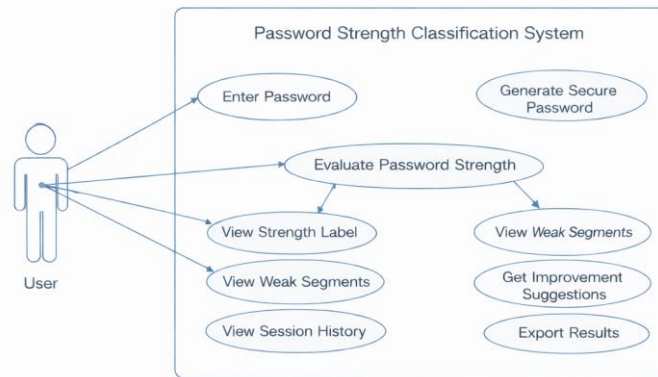


Figure 4.3: Use Case Diagram of Password Strength Classification System

Figure 4.3 illustrates the use case diagram of the Password Strength Classification System. The user interacts with the system to enter passwords, evaluate their strength, view results, receive improvement suggestions, generate secure passwords, and export analysis results.

5. IMPLEMENTATION

This chapter explains the practical implementation of the Password Strength Classification System. It describes how different modules of the system were developed and integrated to evaluate password strength effectively. The system is implemented using Python programming language with a graphical user interface built using Tkinter.

5.1 MODULE DESCRIPTION

The Password Strength Classification System is implemented using a modular approach to ensure clarity, maintainability, and efficient execution. Each module performs a specific task in the overall password evaluation process. The system is divided into the following main modules:

- **User Input Module:** This module is responsible for accepting the password entered by the user through the graphical user interface (GUI). The password is captured using an input field where the user can type the desired password. Basic validation is performed to ensure that the input is not empty before proceeding with further analysis. This module acts as the entry point of the system.
- **Password Analysis Module:** The Password Analysis Module examines the structure of the entered password. It performs checks such as password length validation and character analysis. The module identifies the presence of lowercase letters, uppercase letters, numbers, and special characters. It also detects weak patterns like alphabetical sequences, common words, or repetitive characters that may reduce password security.
- **Strength Evaluation Module:** This module evaluates the overall strength of the password based on predefined rules and calculations. It computes the entropy value of the password to measure its randomness. Using the entropy and assumed attacker speed, the module estimates the approximate

time required to crack the password. Based on these values, the password is classified into categories such as Weak, Moderate, or Strong.

- **Suggestion and Secure Password Generation Module:** This module provides improvement suggestions to help users strengthen weak passwords. Suggestions may include increasing password length, adding uppercase letters, numbers, or special symbols. The module also generates a secure password automatically by combining different character sets, offering users a strong alternative password.
- **Session History and Export Module:** The Session History module stores previously analyzed passwords in a masked format to protect user privacy. It records details such as password strength score and entropy value. This module also allows users to export the session history in CSV format, enabling further analysis or record maintenance.

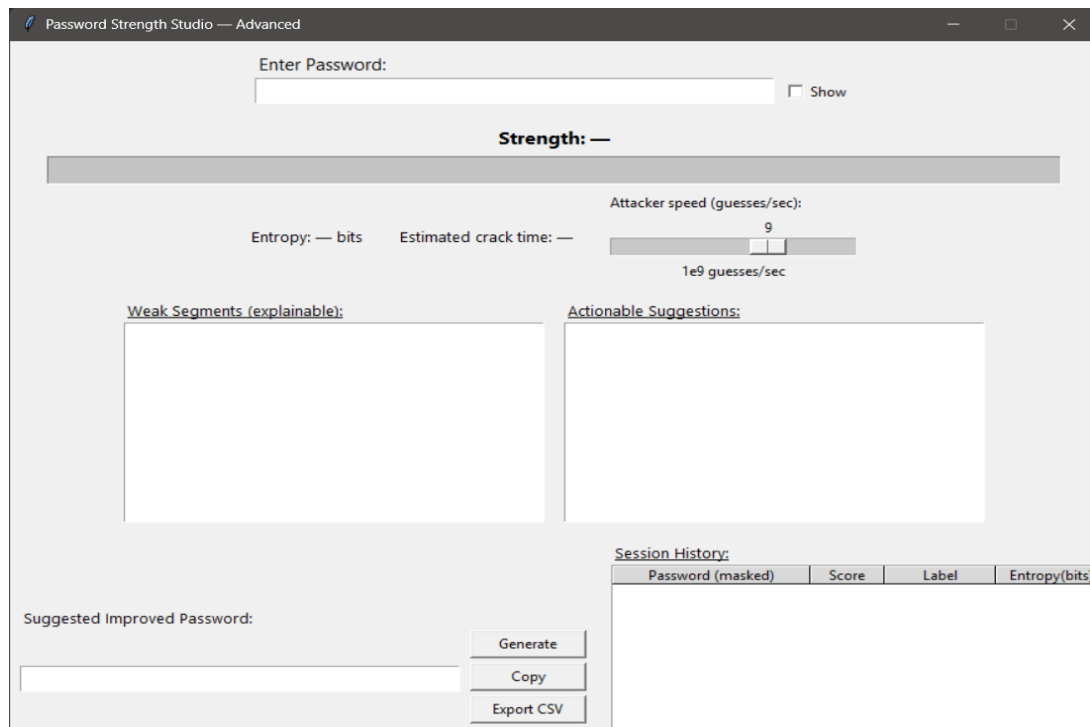


Figure 5.1: GUI – Password Strength Studio Main Interface

Figure 5.1 shows the main graphical user interface of the Password Strength Studio, including the password input field and layout of analysis components.

5.2 GUI DESIGN

The Graphical User Interface (GUI) of the Password Strength Classification System is designed to be simple, interactive, and user-friendly. The interface is developed using the Tkinter library in Python, ensuring ease of use while clearly presenting password strength analysis results.

The main window consists of an input field where users can enter a password for evaluation. A “Show Password” option is provided to allow users to toggle password visibility. Once a password is entered, the system dynamically displays the password strength using a visual strength bar along with a numerical score and classification label.

The GUI also displays important analytical details such as entropy value, estimated crack time, and adjustable attacker speed using a slider. Separate sections are provided to highlight weak password segments and to display actionable suggestions for improving password strength.

Additionally, the interface includes a secure password generation feature that suggests a strong alternative password. Users can generate, copy, and export results easily. A session history table is displayed to store previously evaluated passwords in masked format, ensuring privacy while maintaining usability.

Overall, the GUI design enhances user interaction by presenting complex password analysis results in a clear, structured, and visually understandable manner.

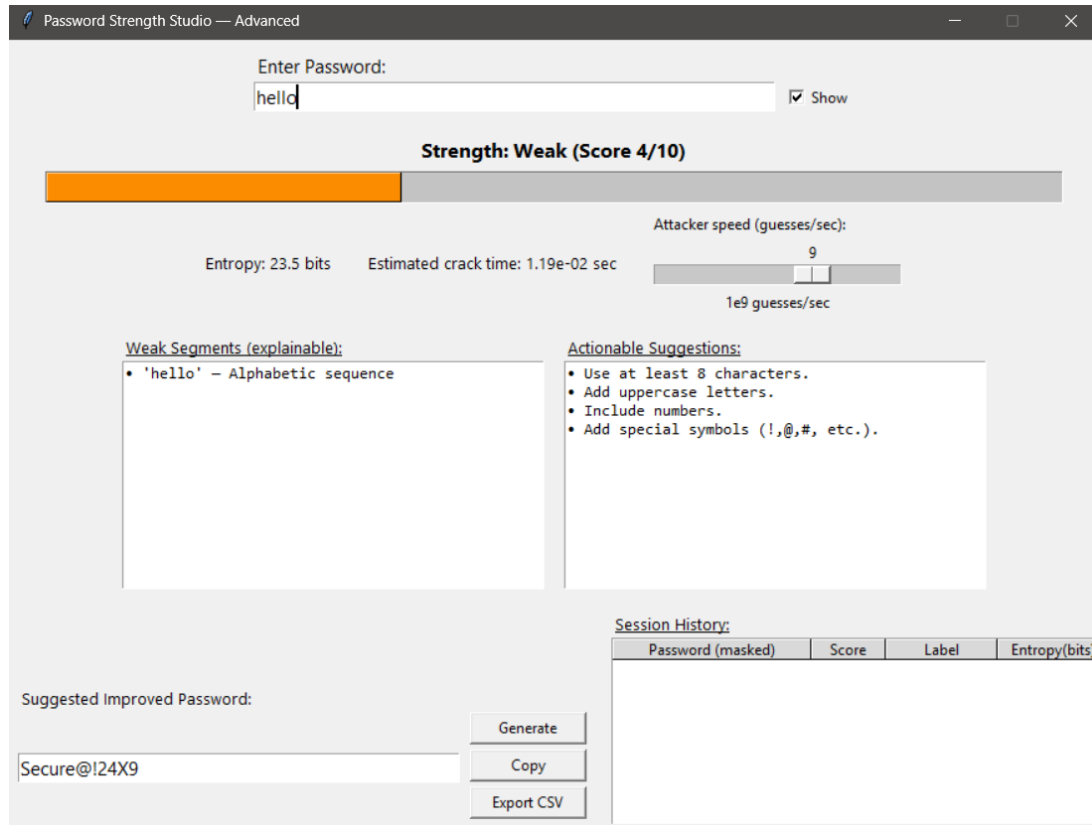


Figure 5.2: GUI Showing Weak Segments and Suggestions

Figure 5.2 illustrates the GUI displaying weak password segments and improvement suggestions after password evaluation.

6. TESTING AND RESULTS

Testing is an essential phase of the Password Strength Classification System, as it ensures that the application functions correctly and produces accurate results under different input conditions. The system was tested using various sample passwords with different combinations of characters such as lowercase letters, uppercase letters, numbers, and special symbols.

The testing process focused on verifying password strength classification, entropy calculation, crack time estimation, detection of weak segments, and generation of improvement suggestions. The results obtained during testing demonstrate that the system successfully evaluates password security and provides meaningful feedback to users.

Test Cases and Results:

This section presents the test cases used to validate the functionality of the system. Each test case includes a sample password, expected outcome, and the actual result produced by the system.

Table 6.1: Test Cases and Results

Password	Entropy (bits)	Estimated Crack Time	Strength Label	Weak Segments Detected
hello	23.5	0.01 sec	Weak	Alphabetic sequence
hello123	41.4	47.02 min	Medium	Alphabetic & numeric sequence
Hello@123	59.0	18.17 years	Strong	Alphabetic & numeric sequence
Security@193X#89	104.9	Several years	Strong	Alphabetic & numeric sequence

Observation:

All test cases produced the expected output. The system correctly classified passwords based on their complexity and character composition.

Sample Password Analysis with Entropy and Crack Time:

This section explains how the system analyzes passwords by calculating entropy and estimating crack time. Entropy measures the randomness of a password, while crack time estimates how long it would take for an attacker to break the password based on a given attack speed.

The system dynamically updates these values as the user enters or modifies a password, allowing real-time evaluation of password strength.

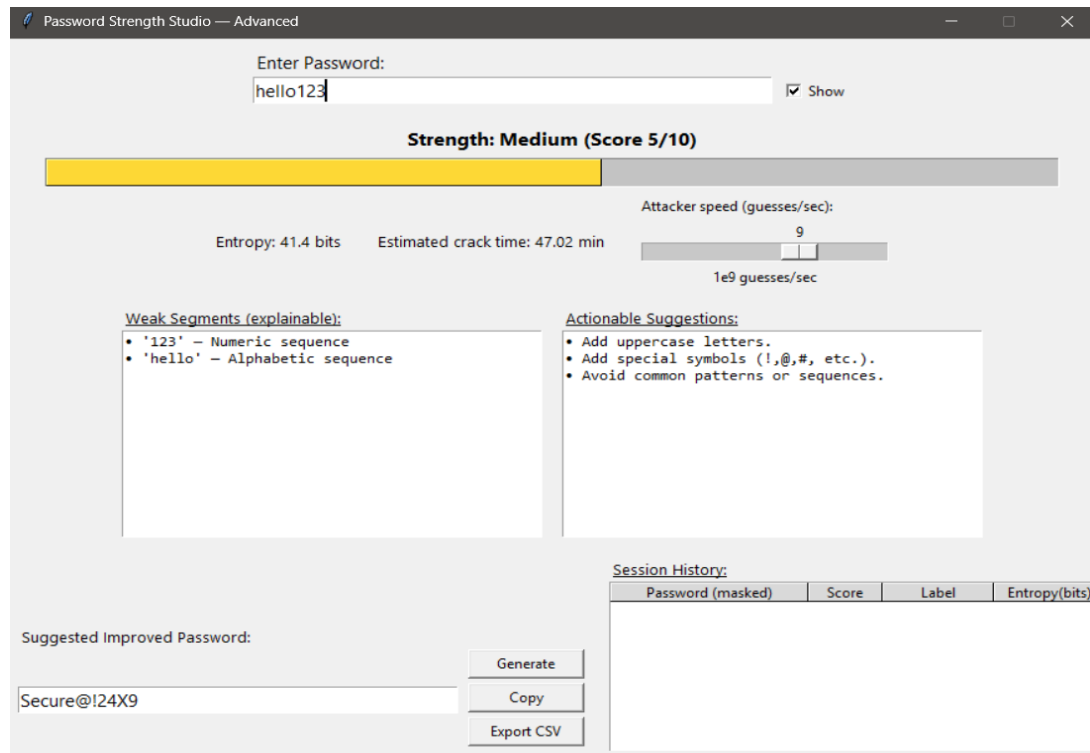


Figure 6.1: Sample Password Analysis Showing Entropy and Crack Time

Figure 6.1 illustrates the output of the Password Strength Classification System when a sample password is entered. The system evaluates the password and displays its strength level, entropy value, and estimated crack time. It also identifies weak segments present in the password and provides actionable suggestions to improve password security. This demonstrates the effectiveness of the system in analyzing passwords and guiding users towards creating stronger passwords.

Table 6.2: Sample Password Analysis with Entropy and Crack Time

Password	Entropy (bits)	Estimated Crack Time	Strength Label	Weak Segments Detected
hello	23.5	0.01 sec	Weak	Alphabetic sequence
hello123	41.4	47.02 min	Medium	Alphabetic & numeric sequence
Hello@123	59.0	18.17 years	Strong	None
Security@193X#89	104.9	Several years	Strong	None

Observation:

As password complexity increases, entropy and crack time also increase significantly. Passwords containing mixed character sets and special symbols provide stronger security and are harder to crack.

Conclusion of Testing:

The testing results confirm that the Password Strength Classification System operates accurately and reliably. The system effectively detects weak passwords, calculates entropy, estimates crack time, and provides useful suggestions for improving password strength. The graphical interface enhances usability by presenting results in a clear and understandable manner.

7. CONCLUSION AND FUTURE SCOPE

Conclusion:

The Password Strength Classification System successfully analyzes user-entered passwords and evaluates their security based on factors such as length, character composition, entropy, and estimated crack time. The system effectively identifies weak password patterns and provides clear strength labels along with actionable improvement suggestions. By presenting real-time feedback through an intuitive graphical user interface, the application helps users understand the importance of strong passwords and encourages better security practices. The modular design of the system ensures accuracy, efficiency, and ease of maintenance, making it a reliable tool for password security assessment.

Future Scope:

The system can be further enhanced by integrating advanced password analysis techniques such as machine learning-based strength prediction and real-world breach database checks. Support for multi-language interfaces and mobile platforms can improve accessibility for a wider user base. Additionally, features such as cloud storage for session history, role-based access, and integration with authentication systems can be implemented. Incorporating real-time security alerts and compliance with modern security standards would further strengthen the system and make it suitable for large-scale applications.

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APPENDIX

APPENDIX – A

Source Code Snippets

```
def calculate_entropy(password):  
    charset = 0  
    if any(c.islower() for c in password):  
        charset += 26  
    if any(c.isupper() for c in password):  
        charset += 26  
    if any(c.isdigit() for c in password):  
        charset += 10  
    if any(c in "!@#$%^&*()" for c in password):  
        charset += 10  
    return len(password) * math.log2(charset)
```

APPENDIX – B

User Manual

Steps to Use the Password Strength Classification System:

1. Launch the application using Python.
2. Enter the desired password in the input field.
3. Click or type to analyze the password.
4. View the password strength, entropy, and estimated crack time.
5. Review weak segments and improvement suggestions.
6. Generate a secure password if required.
7. Export the session history using the “Export CSV” option.

APPENDIX – C

Sample Test Data and Results

Password	Strength	Entropy (bits)	Crack Time
hello	Weak	23.5	0.01 sec
hello123	Medium	41.4	47 min
Secure@124X9	Strong	78.6	Years